STRATEGIC BIOSECURITY PRODUCT PROGRAM

Product Requirement Document (PRD)

Project BioWatch Horizon

Predictive dual-use biotechnology threat monitoring for defensive risk mitigation

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Executive snapshot. BioWatch Horizon is a defensive, analyst-facing intelligence dashboard. It tracks *who has what biomanufacturing capability and where*, screens *documents and data for dual-use red flags* without ever generating or optimizing hazardous content, and models *geopolitical risk scenarios at the policy and capacity level*. Its guardrails are architectural, not cosmetic: the system is built so that offensive uplift is not a feature that can be switched on.

1. Product Overview and Vision

1.1 Product Name

BioWatch Horizon (internal program code: BWH)

1.2 Mission Statement

> **Mission:** To give geopolitical analysts, intelligence agencies, and global health monitors a predictive, AI-powered dashboard that tracks dual-use biotechnology threats and proliferation risks, so that oversight, verification, and countermeasure investment can keep pace with the technology, while protecting legitimate science and global health equity.

1.3 Vision

Within three years, BioWatch Horizon is the common operating picture for dual-use biotechnology governance: a shared, auditable view of global capability, a trusted screening assistant for analysts, and a decision-support tool that shows policy makers where oversight gaps are widest and which interventions close them fastest.

1.4 Product Principles

- **Defense only.** The product detects, contextualizes, and recommends mitigations. It never designs, optimizes, or advises on producing hazardous agents.
- **Equity-aware.** Initiatives such as the WHO mRNA Technology Transfer Hub (South Africa) are treated as legitimate global-health infrastructure. The product surfaces *governance and screening gaps* and supports capacity-building; it does not treat hub partners as adversaries.
- **Evidence over alarm.** Every risk score is traceable to sources, shows its uncertainty, and distinguishes capability from intent.
- **Human accountable.** Analysts and designated officers make decisions; the AI assists. No automated enforcement or accusation.
- **Least exposure.** Outputs are tiered and redacted by design; sensitive detail is not stored or surfaced when a risk tier is enough.

1.5 Scope Summary

In scope (v1.x)	Explicitly out of scope
Capability and infrastructure mapping from open and licensed sources; document and data screening against curated hazard references; strategic scenario and risk-index modeling; audit, compliance, and reporting tooling.	Any generation, redesign, optimization, or troubleshooting of biological agents or production methods; offensive planning of any kind; individual-level surveillance of researchers; automated enforcement actions; operational targeting.

2. The Why: Problem Statement and Justification

2.1 Why Now?

Historically, offensive biological weapons programs demanded state-scale capital, specialized facilities, and rare expertise, a natural barrier to proliferation. The source case study documents how that barrier is eroding through three converging trends:

- **Democratization of mRNA technology.** After COVID-19, mRNA and lipid-nanoparticle (LNP) platforms were deliberately decentralized for health equity. The WHO hub in South Africa (Afrigen Biologics and Biovac) supports 15+ recipient nations. mRNA is a pathogen-agnostic platform: production can be re-pointed by changing the digital sequence input, with source-cited switching times of days to weeks.
- **Automated DNA synthesis.** Benchtop and cloud-lab synthesis allows sequence generation without physical custody of source strains, shrinking the footprint from multi-acre complexes to compact units.
- **AI-driven biological design.** Protein-design and structure-prediction tools lower the expertise needed for biological engineering and raise the stakes of unscreened digital sequence exchange.
- **Regulatory lag.** The 1972 Biological Weapons Convention (BWC) has no physical verification protocol (source: 'Zero BWC physical verification protocols'). It was designed for state stockpiles and large facilities, not modular nodes and digital sequence information.

2.2 The Threat Gap

Existing frameworks largely monitor the **physical shipment of known pathogens** and large, visible facilities. The modern gap sits elsewhere:

Dimension	Legacy oversight assumes	Reality BioWatch Horizon addresses
Unit of control	Physical agents and strains in transit	Digital genetic sequences and design files moving across borders
Facility visibility	Large fixed plants, observable growth capacity	Compact, cleanroom-contained modular units with low observable signature (cell-free production)
Switching time	Months to years (cell-line validation)	Days to weeks by changing a digital template (source-derived)
Verification	Physical inspection	High digital dependency; batches hard to distinguish
Actor set	A few states with industrial programs	Many legitimate hubs, commercial providers, and cloud labs, unevenly equipped with sequence screening

Future-looking concerns identified in the source report, such as stealthy long-incubation agents, demographically targeted agents, agricultural and economic sabotage, and gray-zone deniability, are treated here strictly as **strategic planning categories** for oversight prioritization. The product models *exposure and governance gaps* against these categories, never agent characteristics or methods.

2.3 Target User Personas

Persona	Primary goals	Pain points	What BioWatch Horizon provides
Biosecurity Intelligence Analyst (national agency, defense, or think tank)	Maintain current picture of dual-use capability; triage emerging concerns; brief leadership.	Fragmented open-source data; manual screening of long reports; no common risk language; alert fatigue from false positives.	Infrastructure tracker with provenance; AI-assisted document and sequence-data triage with explainable flags; configurable alerts; brief-ready exports.

Persona	Primary goals	Pain points	What BioWatch Horizon provides
Global Health Policy Maker (ministry, WHO-linked body, funder)	Expand equitable access to biomanufacturing without creating unmanaged risk; target biosecurity funding.	No neutral way to show that a hub is well-governed; risk of equity programs being stigmatized; unclear screening standards.	Governance-readiness views per hub, gap-closing recommendations (screening, KYC, training), and scenario comparisons of policy options.
International Arms Control Inspector (BWC-adjacent, OPCW-style technical staff)	Prepare verification missions and design inspection regimes for modular facilities.	No standardized facility signatures; inspections rely on physical visibility that no longer exists; limited digital-record audit tooling.	Inspection-prioritization lists, compliance-document audit assistant, evidence chain-of-custody logs, and mission-planning packs.

3. Core User Stories and Feature Scope

Priority key: **P0** = launch-blocking, **P1** = required for GA, **P2** = post-GA.

Feature 1: Global Dual-Use Infrastructure Tracker

An analytical dashboard and interactive map monitoring global mRNA hubs, regional bioreactor and LNP-formulation capacity, and commercial DNA synthesis providers.

ID	As a...	I want to...	So that...	Pri.
F1-01	Analyst	view an interactive map of mRNA hubs, synthesis providers, and cloud labs with filters for region, platform type, and governance status	I can see capability concentrations and blind spots at a glance	P0
F1-02	Analyst	open an entity profile showing capacity class, ownership, partners, and a source-linked evidence trail with confidence ratings	I can defend every data point in a briefing	P0
F1-03	Policy maker	see a governance-readiness checklist per facility (sequence screening, customer verification, biosafety accreditation, incident reporting)	I can target support to close gaps without stigmatizing the facility	P1
F1-04	Inspector	generate a ranked inspection-priority list from capability and governance indicators	I can plan limited verification resources	P1
F1-05	Analyst	receive alerts when a tracked entity changes (new capacity, new partnerships, regulatory action)	I stay current without manual monitoring	P1
F1-06	Any user	compare regions over time using trend charts of capacity and screening coverage	I can spot proliferation trajectories early	P2

Functional requirements

- Ingest from open-source registries, regulatory filings, trade and procurement data, academic publications, and licensed datasets; every record stores source, retrieval date, and confidence grade.
- Capacity is shown as **classes** (for example small, medium, large; modular or fixed), not engineering parameters.
- Entity resolution and deduplication across sources; analyst-editable annotations with versioned history.
- Role-based map layers; classified overlays are only visible to cleared roles.
- Export to briefing formats (PDF, slide-ready images) with automatic classification banners.

Acceptance criteria

- Map loads under 3 seconds for 5,000 entities on standard analyst hardware.
- 100% of displayed attributes link to at least one source; unsourced values cannot be published to shared views.
- Governance-readiness status always shows 'unknown' rather than 'compliant' when evidence is absent.

Feature 2: AI-Powered Sequence Risk Evaluator

A natural-language interface where an analyst uploads text, reports, or data files containing genetic sequence information. The system flags dual-use anomalies and potential hazardous homologies by **screening against curated reference lists** and returns a **risk tier with rationale and escalation path**. It is a screening and triage tool, not a design or analysis workbench.

Design constraint: Feature 2 is a *classifier and router*. It never outputs modified, improved, or alternative sequences, never explains how to make a flagged item more potent, transmissible, stable, or evasive, and never returns the matched hazardous reference content itself. Output is limited to tier, confidence, category of concern, provenance, and recommended next step.

ID	As a...	I want to...	So that...	Pri.
F2-01	Analyst	upload a report, order log, or sequence data file and ask in plain language whether anything warrants escalation	I can triage large volumes quickly	P0
F2-02	Analyst	see a tiered result (Clear / Review / Escalate) with confidence, category of concern, and the document locations that triggered it	I can verify the flag and avoid wasted effort on false positives	P0
F2-03	Inspector	screen a facility's order and customer-verification records for anomalous patterns (for example split orders, mismatched end-use statements)	I can focus inspection on genuine compliance risks	P1
F2-04	Policy maker	run a screening-coverage audit of a provider's published procedures against a recognized standard	I can assess readiness without handling sequence data	P1
F2-05	Analyst	route 'Escalate' results to a designated biosecurity officer with a tamper-evident case record	Handling follows protocol and is auditable	P0
F2-06	Analyst	give feedback on a flag (true positive, false positive, unclear)	The screening models are tuned against analyst judgment	P1

Functional requirements

- Screening runs against curated, access-controlled reference sets maintained by the biosecurity data steward; references are never exposed to end users or to the LLM prompt in raw form.
- A deterministic homology and rules engine produces the primary determination; the LLM handles document parsing, context extraction, and explanation text only.
- Escalate-tier items trigger a quarantine workflow: content is encrypted, access-logged, and visible only to designated officers.
- Language interface supports analyst questions about *what was flagged and why*; any request to modify, optimize, or generate sequences is refused and logged (see Section 5).
- Support for text, PDF, spreadsheet, FASTA-style, and structured order-log formats with virus and malicious-file scanning on upload.

Acceptance criteria

- Zero responses containing generated, edited, or optimized sequences or production guidance during red-team test suites (hard release gate).
- Every result includes tier, confidence interval, trigger locations, and a next-step recommendation.
- Analyst feedback loop live before GA; weekly precision and recall reporting.

Feature 3: Geopolitical Threat Horizon Predictor

An intelligence module using large language models and structured quantitative models to run **strategic, policy-level** gray-zone scenarios and compute risk indexes based on regional instability, technology access, and governance maturity.

Design constraint: Scenarios are built from *actors, capacities, governance variables, and political events* (for example export-control changes, funding shifts, a hub losing a partner, a disputed outbreak attribution). They never include agent selection, engineering approaches, or operational attack planning. The output is a risk index, indicator watchlist, and mitigation options.

ID	As a...	I want to...	So that...	Pri.
F3-01	Analyst	view a Regional Dual-Use Risk Index decomposed into instability, technology access, governance maturity, and verification gap sub-scores	I understand what drives each score	P0
F3-02	Policy maker	run what-if scenarios on policy options (for example mandatory screening, funding for oversight, export-control changes) and compare index outcomes	I can choose interventions with the largest risk reduction	P1
F3-03	Analyst	generate gray-zone attribution-ambiguity scenarios (for example an unexplained regional outbreak with disputed origin) focused on decision-making, response time, and communications readiness	I can stress-test institutional readiness	P1
F3-04	Inspector	see leading indicators and early-warning watchlists linked to the index	I know which observable events should trigger review	P1
F3-05	Executive	export a briefing with assumptions, uncertainty ranges, and dissenting model views	Decision makers see confidence and limits, not false precision	P0
F3-06	Analyst	back-test the index against historical events and publish calibration results	I can trust, or challenge, the model	P2

Functional requirements

- Index = transparent weighted model over sourced indicators (conflict and instability data, export-control and trade data, capacity classes from Feature 1, governance-readiness from Feature 1 and 2). Weights and formulas are inspectable.
- LLM role: scenario narrative drafting, red-team style 'what might we be missing' prompts about institutional response, and plain-language explanation. It does not set scores.
- Every scenario output includes assumptions, uncertainty bands, and a statement that scenarios are not predictions.
- Scenario templates are curated and versioned by the ethics board; free-form scenario prompts pass through the same guardrail layer as Feature 2.

Acceptance criteria

- Index sub-scores reproducible from stored inputs; any score can be re-derived on audit.
- Calibration report (Brier score or equivalent) published before each major model release.
- Scenario red-team suite shows no operational or technical-uplift content across all templates.

3.4 Cross-Cutting Capabilities

- **Case management:** tamper-evident records for escalations, with chain-of-custody and legal-hold support.
- **Collaboration:** shared workspaces with compartmentalized access and need-to-know controls.
- **Reporting:** classification-aware exports and automated audit packs for oversight bodies.

4. Tech Stack and System Architecture

4.1 Frontend and Analytics UI

A clean, high-density dashboard designed for long analyst sessions: dark and light themes, keyboard-first navigation, and WCAG 2.2 AA accessibility.

Layer	Choice (recommended)	Rationale
Framework	React with TypeScript, component library with design tokens	Strong ecosystem, type safety for complex state
Geospatial	MapLibre GL with vector tiles; deck.gl for dense overlays	Open-source, performant with thousands of entities, offline-capable for accredited networks
Charts	D3 and Apache ECharts	Dense small multiples, time series, network graphs, uncertainty bands
Layout	Multi-pane analyst workspace: map, entity panel, evidence timeline, case queue	Reduces context switching; supports side-by-side comparison
State and data	TanStack Query, WebSocket for alerts	Efficient caching and real-time alerting
Visual rules	Classification banner on every view; uncertainty always visible; 'unknown' is a first-class state	Prevents over-reading of sparse data

4.2 Core Intelligence Layer

State-of-the-art large language models are accessed through secure, enterprise-grade APIs for unstructured intelligence text parsing, scenario generation, and compliance-document auditing. The LLM is deliberately placed **behind** deterministic controls rather than in front of them.

Component	Responsibility
Ingestion and parsing	Extract entities, dates, organizations, and claims from reports and filings; produce structured records with source spans.
Screening engine (deterministic)	Rules and homology checks against access-controlled reference sets; primary basis for Feature 2 tiers.
Guardrail gateway	Input and output classifiers, policy engine, refusal logic, and logging; all LLM traffic passes through it (see Section 5).
LLM orchestration	Prompt templates versioned and reviewed; retrieval-augmented generation limited to approved corpora; no tool access to raw reference sets.
Risk index service	Transparent statistical model with sourced indicators; produces sub-scores, uncertainty, and explanations.
Scenario service	Template-driven scenario builder using policy-level variables; LLM drafts narratives only.
Audit and observability	Immutable logs of prompts, outputs, refusals, overrides, and user actions; anomaly detection on user behavior.

4.3 Platform and Security Architecture

Area	Specification
Hosting	Sovereign or government-accredited cloud (for example FedRAMP High or national equivalent) with an air-gapped deployment option; model access via private endpoints

Area	Specification
Data stores	PostgreSQL with PostGIS (entities, geospatial); graph database for ownership and partnership networks; object store with envelope encryption for documents; vector index restricted to approved corpora
Identity and access	SSO with hardware-key MFA; attribute- and role-based access control; compartmented need-to-know groups; just-in-time privilege elevation
Data protection	Encryption in transit (TLS 1.3) and at rest (customer-managed keys); field-level encryption for escalated content; data-loss prevention on exports
LLM data handling	Zero-retention API terms; no training on customer data; prompts and outputs logged only in the customer's audit store
Resilience	99.9% availability target; multi-zone deployment; tested disaster recovery with RTO under 4 hours
Assurance	Annual independent penetration test; continuous red-teaming of the guardrail layer; supply-chain (SBOM) controls

5. Success Metrics, Safety Guardrails, and Ethics

5.1 Key Performance Indicators

KPI	Definition	Target (GA)	Measurement
Threat detection recall (Feature 2)	Share of expert-confirmed dual-use cases flagged Review or Escalate	>= 95% on held-out benchmark	Quarterly blind evaluation by independent panel
Precision at Escalate tier	Share of Escalate flags confirmed by officer review	>= 80%	Officer disposition records
False positive rate (Review + Escalate)	Share of benign items incorrectly flagged	<= 5% overall; <= 1% at Escalate	Benchmark plus live feedback
Time to triage	Median time from upload to analyst decision	< 15 minutes (vs. multi-hour manual baseline)	Workflow telemetry
Infrastructure data coverage	Share of known mRNA hubs and synthesis providers with sourced, current profiles	>= 90% within 12 months	Registry reconciliation
Data freshness	Median age of entity record updates	< 30 days	Pipeline monitoring
Index calibration	Calibration error of Risk Index on back-tests	Published, with year-over-year improvement	Back-test report
Guardrail integrity	Policy-violating outputs in red-team suite	0 (hard release gate)	Continuous red-team harness
Analyst adoption and trust	Weekly active analysts; trust score on survey	>= 70% of licensed users; >= 4.0/5	Usage analytics and survey

Targets are initial proposals and must be validated against pilot data. Threshold tuning must weigh the asymmetric costs of misses (security) and false alarms (stigma toward legitimate research and equity programs).

5.2 Strict Alignment Boundaries

The AI integration must **strictly block** any query that attempts to reverse-engineer, weaponize, or optimize dangerous biological agents. The product's purpose is exclusively defensive risk mitigation and compliance monitoring.

Blocked request category	Required system behavior
Designing, modifying, enhancing, or optimizing any biological agent or toxin, including transmissibility, lethality, stability, immune or detection evasion, or host or demographic targeting	Refuse; log; no partial answer; notify security officer on repeated attempts
Production, scale-up, formulation, delivery, or dissemination methods for hazardous agents; troubleshooting of such processes	Refuse; log
Requests to generate, complete, 'fix', or rewrite genetic sequences associated with hazardous agents	Refuse; return only screening tier if a file was submitted
Requests to evade or defeat sequence screening, customer verification, export controls, or inspection regimes	Refuse; log; flag account for review
Operational attack planning, target selection, or optimization of a scenario for maximum harm	Refuse; scenario engine accepts governance and policy variables only
Attempts to extract the contents of curated hazard reference sets or guardrail rules (prompt injection, role-play, encoding tricks)	Refuse; log; raise security alert

Layered enforcement

- **Architectural:** the LLM has no access to raw hazard references and no sequence-generation tools; it cannot output what it cannot reach.
- **Input classifier:** screens prompts and uploads for the blocked categories before reaching the model.
- **Output classifier:** screens every response for technical-uplift content and sequence-like strings; responses failing the check are replaced with a safe summary.
- **Template-bound scenarios:** Feature 3 uses approved policy-level templates; free-form inputs are normalized into the same variable space.
- **Adversarial testing:** continuous internal and external red-teaming, plus a bug-bounty style disclosure program for guardrail bypasses; any bypass is a P0 incident.
- **Misuse detection:** repeated refusals, probing patterns, or unusual data exports trigger insider-risk review.

5.3 Governance and Ethics

Topic	Commitment
Oversight body	Independent Ethics and Safety Board (biosecurity, international law, global-health equity, civil liberties) with veto over launch and major releases
Human accountability	No automated accusations or enforcement; every Escalate decision is made by a named officer and recorded
Equity and non-stigmatization	Hub and facility views use language of governance readiness and support needs. Public naming of specific legitimate facilities as threats requires board-level review. Features provide capacity-building recommendations, not only warnings.
Bias and fairness	Regional risk indexes are tested for geographic and political bias; model cards document limits; geographic factors are never treated as evidence of intent
Privacy and civil liberties	No individual-level scientist surveillance; personal data minimized; retention limits and legal-basis review per jurisdiction
Transparency	Model cards, index methodology, and guardrail policy summaries available to accredited oversight bodies; public methodology note
Information hazard management	Outputs use tiered detail; sensitive findings are shared by need-to-know and never copied to unmanaged systems
Legal compliance	Review against export-control, data-protection, and international humanitarian and treaty obligations in each deployment jurisdiction
Incident response	Defined process for guardrail failures, data exposure, and misuse, with notification timelines and independent post-incident review

6. Delivery Roadmap, Risks, and Open Questions

6.1 Phased Roadmap

Phase	Timeline	Scope	Exit gate
0. Foundations	Months 0-3	Ethics board charter; threat model; data-source agreements; guardrail architecture and red-team harness	Board approval; guardrail design review passed
1. Tracker MVP	Months 3-7	Feature 1 core (map, entity profiles, evidence trail, alerts); SSO and audit logging	Pilot with 2 partner agencies; data coverage >= 60%
2. Screening Beta	Months 7-12	Feature 2 triage with deterministic engine, tiers, case management; analyst feedback loop	Zero guardrail failures in red-team gate; recall and precision targets on benchmark
3. Horizon Beta	Months 12-17	Feature 3 risk index, policy what-if, scenario templates; calibration reporting	Independent methodology review; back-test report
4. General Availability	Months 17-20	Hardening, accreditation, multi-tenant compartmenting, international partner onboarding	Accreditation granted; KPI targets met in pilots

6.2 Key Risks and Mitigations

Risk	Impact	Likelihood	Mitigation
Guardrail bypass leads to harmful output	Critical	Low-Med	Architectural isolation, layered classifiers, continuous red-team, P0 incident process, kill-switch
Product itself becomes a high-value target (insider or external)	Critical	Medium	Compartmentation, hardware-key MFA, DLP, insider-risk monitoring, sovereign hosting
Stigmatization of legitimate hubs and equity programs	High	Medium	Governance-readiness framing, board review of naming, capacity-building outputs, engagement with hub operators
Overreliance on risk scores (false precision)	High	Medium	Uncertainty shown by default, dissent views, mandatory assumptions, analyst training
Sparse or biased source data skews index	High	High	Confidence grades, 'unknown' state, bias audits, multiple-source requirement
High false positives erode trust and burden researchers	Medium	Medium	Tiering, feedback tuning, precision targets, appeal and correction process
Legal and export-control complexity across jurisdictions	Medium	High	Jurisdiction-specific deployment configurations, early legal review, regional data residency
LLM vendor change or model behavior drift	Medium	Medium	Model-agnostic orchestration, regression suites, pinned versions, deterministic core

6.3 Open Questions for Stakeholders

- Which agencies and international bodies are eligible for each access tier, and who adjudicates disputes about access?
- Who owns and maintains the curated hazard reference sets, and under what governance and update cadence?
- What disclosure and engagement model applies to facilities flagged as governance gaps (for example advance notice or support offers)?
- Should a sanitized, unclassified edition of the infrastructure tracker be offered to global-health stakeholders to support transparency and trust?

- What is the minimum acceptable recall at Escalate tier given the asymmetric cost of misses versus false alarms?

Appendix A. Glossary

Term	Meaning
BWC	Biological Weapons Convention (1972); prohibits development and stockpiling of biological weapons; lacks a formal verification regime
Dual-use	Technology, knowledge, or infrastructure usable for both beneficial and harmful purposes
DSI	Digital Sequence Information: genetic sequence data held and exchanged electronically
LNP	Lipid nanoparticle; delivery vehicle for mRNA
mRNA hub	Facility or network transferring or hosting mRNA vaccine manufacturing capability, for example the WHO mRNA Technology Transfer Hub in South Africa
KYC	Know Your Customer: verification of customer identity and end-use before fulfilling synthesis orders
Gray-zone	Hostile activity below the threshold of overt armed conflict, often with plausible deniability
Homology screening	Comparing submitted sequence data against curated references to detect similarity to hazardous agents
Risk tier	Clear, Review, or Escalate classification returned by the Sequence Risk Evaluator

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